

HW 2 Report 112550151 周佳瑩

Implementation (50%)

- **Code:**

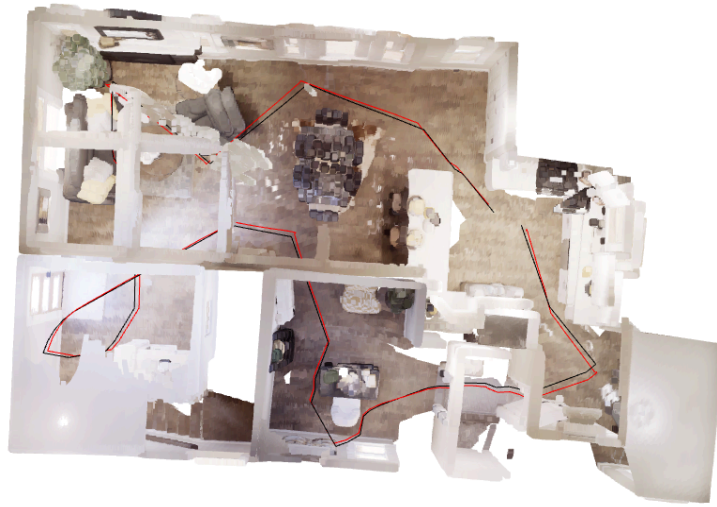
I do unprojection by first converting the inputs, depth, and creating a meshgrid. Then I implement the logic. I calculate the coordinations according to Habitat's cameras, do a validity filtering to filter out any noise, combine the three (H, W) arrays into one (H, W, 3) array, flatten and filter it, and use the provided o3d functions to generate the point cloud. Next, I apply normal estimation on non-empty point clouds to first find the normal of up to 30 neighboring points on the plane with the provided radius to be the normal vector of the point, and use normal orientation to make sure the orientation is consistent over the 30 nearest points.

In the main reconstruction function, some conversion and preprocessing are implemented in the main reconstruction function. For global registration, the initial guess is set as identity, and I implement RANSAC with repeatedly sampling 4 point pairs that are matched by FPFH similarity, computing a transform from those 4 pairs, and counting the inliers. The best transform is found and kept after iterations or 500 consecutive good results. Next, the selected ICP is executed. A threshold is set as the maximum correspondence distance, and I set the voxel size to be 1.5 to be tight enough to avoid wrong matches but loose enough to tolerate small misalignments from RANSAC. There is also an ICP quality gating to validate the result.

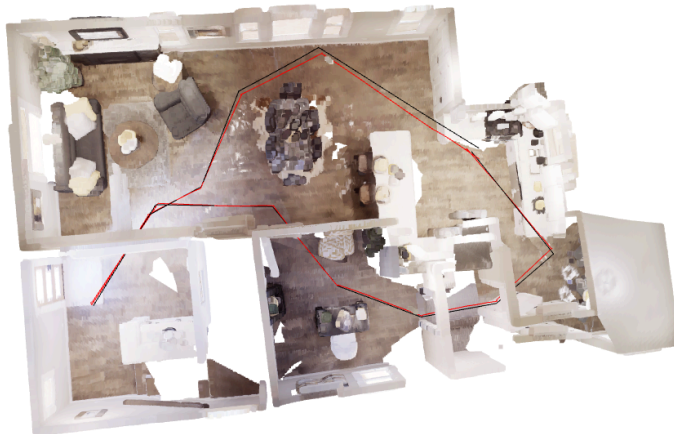
Lastly, I implement the pose update and point accumulation by applying the transformation and merging it into the growing global map, making sure to deep copy the data to avoid modifying the original data. I also remove the ceiling here by finding the camera height and applying a cut off threshold.

- **Result & Discussion:**

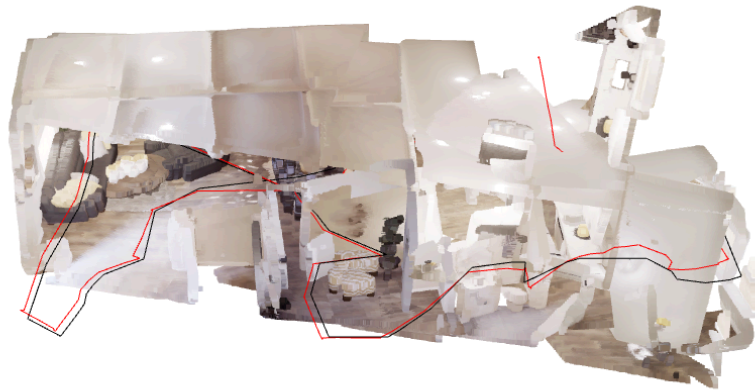
- Screenshots of apartment_0 Floor 1 reconstruction.
Data size = 265



Data size = 180



Data size = 428 (I failed to cut off the ceiling)



- A comparison table that includes:
 - Mean L2 distance between estimated and ground truth camera poses.
 - Total execution time.
 - The hyperparameters you have tested. Discuss the results and share your findings.

I have come to realize that bigger doesn't always mean better. I thought bigger data size meant more data could be captured, but at the same time it also provided more frames to match, which the code struggled to do. A suitable amount or trajectory for collecting data may be more important than the size of data in this case.
 - (Bonus) If you implement your own ICP algorithm, compare your results with the Open3D version and highlight the techniques or tricks you used to improve performance.

ICP	Data Size	Mean L2 Dist	Execution time
Default	180	0.088290m	65.08s
Default	265	0.108171m	97.92s
Default	428	0.346549m	138.82s
My ICP	180	0.778663m	270.79s
My ICP	265	0.646691m	380.14s

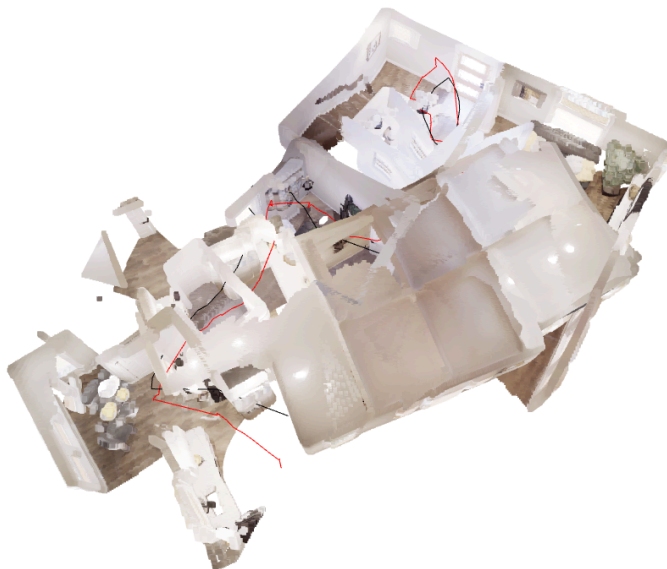
My ICP	428	2.425663m	554.05s
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My ICP

Data size = 180



Data size = 265



○ **Questions (20%):**

- What happens if you perform ICP without Global Registration (RANSAC)? Why?
If ICP is performed without global registration, ICP will

most likely converge to a wrong local minimum, as RANSAC is in charge of providing the initial alignment while ICP finds nearest neighbors based on current positions but not true correspondences, meaning it can only minimize distance between points that are already roughly aligned. If two points are far apart, their nearest neighbors are the geometrically closest target points that might actually be from a different part of the scene. ICP might optimize towards that direction, which is incorrect, without the pointer from RANSAC.

- Describe any tricks used to improve your ICP stability (e.g., Voxel size, Huber loss, etc.).
 1. Two-stage voxel sizing: I used a different voxel for RANSAC and for ICP, with a coarser one for RANSAC for faster and more distinctive feature matching and a finer one for ICP to preserve geometric detail.
 2. RANSAC quality gating: The result of RANSAC is validated against fitness, RMSE, and physically plausible translation magnitude to make sure that the result is good enough to feed into ICP.
 3. Motion prediction baseline: When RANSAC fails, the last known relative transform is used as the initial guess rather than using identity. This is to exploit the fact that the agent moves continuously, so it's likely that the movement is the same or similar.
 4. Point-to-Plane instead of Point-to-Point ICP: This is to minimize distance along surface normals rather than direct Euclidean distance between points.